3643. Flip Square Submatrix Vertically

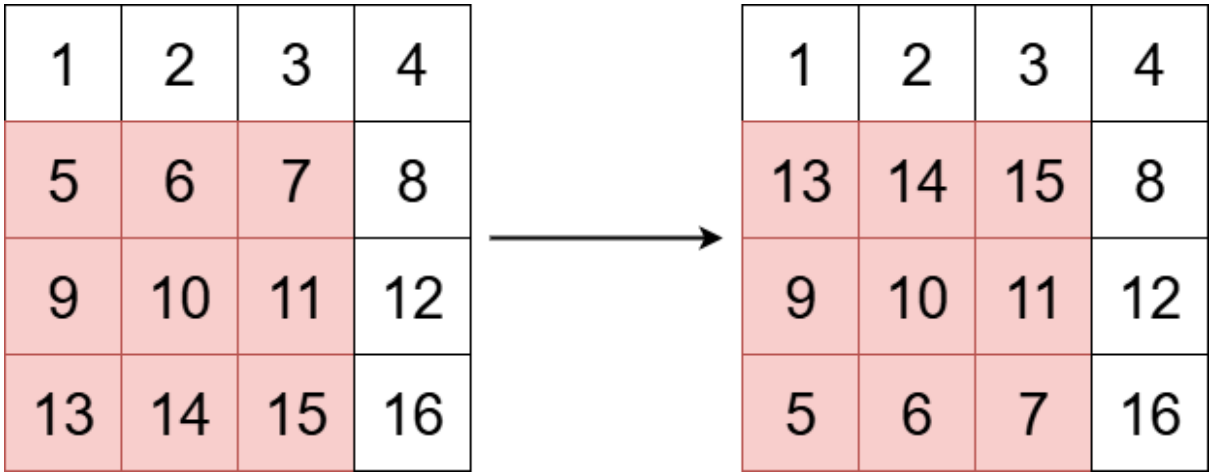
You are given an m x n integer matrix grid, and three integers x, y, and k.

The integers x and y represent the row and column indices of the **top-left** corner of a **square** submatrix and the integer k represents the size (side length) of the square submatrix.

Your task is to flip the submatrix by reversing the order of its rows vertically.

Return the updated matrix.

Example 1:



Input: grid = [[1,2,3,4],[5,6,7,8],[9,10,11,12],[13,14,15,16]], x = 1, y = 0, k = 3

Output: [[1,2,3,4],[13,14,15,8],[9,10,11,12],[5,6,7,16]]

Explanation:

The diagram above shows the grid before and after the transformation.

Example 2:

Input: grid = [[3,4,2,3],[2,3,4,2]], x = 0, y = 2, k = 2

Output: [[3,4,4,2],[2,3,2,3]]

Explanation:

The diagram above shows the grid before and after the transformation.

Constraints:

- m == grid.length
- n == grid[i].length
- 1 <= m, n <= 50
- 1 <= grid[i][j] <= 100
- 0 <= x < m
- 0 <= y < n
- 1 <= k <= min(m - x, n - y)

Solution:--

```
class Solution {
public:
   vector<vector<int>> reverseSubmatrix(vector<vector<int>>& grid, int x,
                                        int y, int k) {
      for (int i0 = x, i1 = x + k - 1; i0 < i1; ++i0, --i1) {
         for (int j = y; j < y + k; ++j) {
            swap(grid[i0][j], grid[i1][j]);
         }
      }
      return grid;
   }
};
```